Operations Decision Memo — Project Parcel

Memo details

To: Operations Head
From: Project team
Subject: Limited pilot of late-shipment risk ranking
Decision requested: Approve a shadow-mode evaluation, not automated interventions.

Recommendation

For this synthetic demonstration, rank shipments by predicted late-delivery risk and send the highest-risk 10% to a human review queue. Do not automatically reroute, upgrade, or message customers based only on the model score.

What the demonstration suggests

On the held-out synthetic test set, the top 10% queue selected 120 of 1,200 shipments. Approximately 65.83% of those selected shipments were late. The queue captured approximately 18.33% of all late shipments. This indicates useful concentration of risk in the queue, but it also means most late shipments were not selected.

Workload and cost

A threshold chosen under illustrative cost assumptions flagged approximately 86.4% of the validation set. That would create a queue far larger than the assumed 10% review capacity. The top-10% policy directly limits workload to 120 cases per 1,200 shipments. No monetary savings or intervention ROI can be claimed because actual intervention costs, missed-late costs, and intervention effectiveness have not been measured.

Proposed operating process

1. Run the model in shadow mode on dispatch-time data.
2. Produce a daily ranked list, limited to the agreed review capacity.
3. Have an operator inspect the flagged parcels and decide whether an intervention is appropriate.
4. Record the reason, action, cost, and eventual delivery outcome.
5. Compare outcomes with a baseline and monitor feature drift, calibration, and late-delivery capture.

What could go wrong

- Inputs may not be available at dispatch or may contain future information.
- Real shipment patterns may differ substantially from the synthetic data.
- The model may miss many late parcels, including severe cases.
- A distribution shift may reduce ranking quality.
- Human reviewers may not have enough information or time to act effectively.

What remains uncertain

The results do not establish real-world accuracy, causal drivers of delay, financial benefit, or the best review capacity. Carrier and planned-load features need point-in-time validation. Actual operating costs and intervention success rates must be collected.

Decision

Approve only a controlled shadow pilot after data access, feature timing, privacy, and monitoring are reviewed. Do not automate customer promises or expensive interventions until prospective results demonstrate reliable benefit.